

GALAXY CLUSTERS

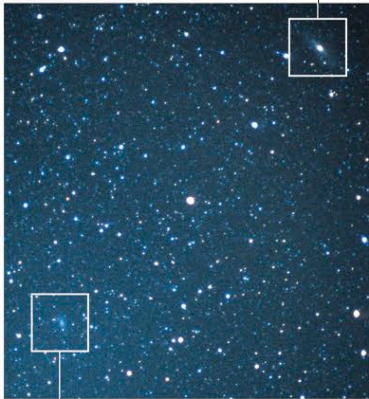
The shape and size of galaxy clusters are thought to be linked to their evolution. Clusters range from small groups comprising young, gas-rich irregular and spiral galaxies to highly evolved clusters dominated by giant ellipticals, with a central cloud of gas so hot that it emits X-rays. Astronomers can study details in nearby clusters that are too faint to see in

DIVERSE CLUSTER

Galaxy cluster Abell S0740 lies more than 450 million light-years away in Centaurus. The center of the cluster is dominated by a giant elliptical galaxy, while spirals and smaller ellipticals orbit farther out.

distant clusters. Earth's neighboring clusters do not offer a spectacle to stargazers, however, because clusters are so vast that their members are widely scattered across the sky. To appreciate clusters in a single picture, it is necessary to peer tens of millions of light-years into deep space.

Andromeda Galaxy, M31



Triangulum Galaxy, M33

LOCAL GROUP MEMBERS

Because Earth is in the midst of the Local Group, the galaxies are scattered around the sky. However, two large members, M33 and M31, are near enough in the sky to appear in the same frame.

IRREGULAR CLUSTER

Local Group



DISTANCE

0–5 million light-years

NUMBER OF GALAXIES

46

BRIGHTEST MEMBERS

Milky Way;
M31 (magnitude 3.5)

ANDROMEDA AND TRIANGULUM

The Local Group is the small galaxy cluster of which the Milky Way is a member. From Earth, its members appear dispersed throughout the sky, but some of its galaxies are grouped in the constellations of Andromeda and Triangulum. In space, the core of the group comprises about 30 members in a region just over 3 million light-years across. It is dominated by the Andromeda Galaxy (M31; see pp.312–313) and the Milky Way. Most of the smaller galaxies orbit close to one or

the other of these large spirals. The third large spiral in the group, M33 (see p.311), may also be trapped in a long orbit around M31.

Outnumbering these spirals is a host of dwarf elliptical and irregular galaxies. Examples include SagDEG and the two Magellanic Clouds (see pp.310–311), as well as M110 and M32, both ellipticals orbiting the M31

spiral. The Local Group appears to be relatively young. Its major galaxies are all spirals, and there is little matter in the space between galaxies—most of the cluster's gas is still trapped in the spirals. It is in an early state of cluster evolution. The Milky Way is currently colliding with the Magellanic Clouds and is heading inexorably toward an ultimate merger with M31.

BARNARD'S GALAXY

This small, irregular galaxy (right), cataloged as NGC 6822, lies 1.7 million light-years away within the Local Group. It is rich in gas and dust, with many pinkish star-birth regions.



FORNAX DWARF GALAXY

This dwarf spheroidal galaxy (left) has no obvious nucleus. Such faint and diffuse galaxies are easily missed in more distant galaxy clusters, but they are probably the most numerous.

THE MILKY WAY GALAXY

A major member of the Local Group is the Milky Way Galaxy. Earth is within the galaxy's disk, so our view is edge-on and stretched across the sky.